

MUHAMMAD RAYED

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EDUCATION

Lahore University of Management Sciences

Bachelor's in Computer Science

2022 - 2026

- **Relevant Coursework:**, Deep Learning, Computer Vision, AI on Edge Devices, AI for Robotics

Beaconhouse School System

O & A-levels – Computer Science, Mathematics, Physics

Lahore, PK | 2017 - 2022

PROFESSIONAL EXPERIENCE

LUMS Computer Vision & Graphics Lab – Research Intern

June 2025 – Ongoing

Animal Detection in Camera trap Images

- Responsible for developing an edge-aware pipeline to improve the currently deployed **object detection** techniques that label rain streaks as animals
- Developed and trained a **U-Net** to detect rain, used a **YOLO** model to detect animals, achieving classification with **IOU scores**
- Achieved **98%** animal and rain **classification accuracy**

Image DeRaining

- Responsible for developing a pipeline to recover information in rain-deteriorated images
- Developed a DeRaining methodology modeling rain streaks as convolution of clear raindrops and motion blur kernels
- Created & Annotated a synthetic dataset of **10,000+** images to engineer and train a CNN based kernel estimator model
- Currently improving the kernel estimator model to achieve the desired metrics

CodeKids PK– Lead Instructor

Lahore, PK | September 2024 – September 2025

- Led and taught over **150** students in on-site **Coding Camps** on programming, machine learning, and software development
- Designed and edited lesson plans based on student behavior and learning habit analyses
- **Stakeholder Communication:** Documented and reported class outcomes to parents, and helped them with queries
- **Tech Stack used:** Python, MachineLearningforKids, Scratch, Thunkable, Minecraft Education

PROJECTS

Deep Learning – LLM Compression and Deployment [Github Link](#)

- Compressed **llama 3.1 8b's** memory footprint by nearly **80%**, going from **16GB** to **3.99GB**
- **Deployed** the model to, and ran inference on an edge device with a RAM of **8GB**, while retaining its performance
- Tech Stack used: **llama.cpp, python, modal, llama 3.1**

Machine Learning - Classification of Urdu News Articles [Github Link](#)

- **Trained** and **tested** 3 machine learning models in **python** to classify Urdu News Articles into 5 categories based on their content only, achieving a highest accuracy of **97.91%**
- Libraries used: **Pytorch, Pandas, Numpy, TensorFlow, Scikit, Matplotlib, Seaborn**
- **Web scraped** 5 Pakistani News Websites, apply **Exploratory Data Analysis(EDA)** to the content collected.

Data Science – Air Crash Data Analysis and Cause Prediction [Github Link](#)

- **Exploratory Data Analysis(EDA):** Analyzed over **100 years** of flight data in Python to find insights and trends
- **Data Cleaning & Feature Selection:** Cleaned a messy dataset and chose relevant features for prediction models
- Built **Machine Learning models** to predict crash causes from summaries and generate a route's risk factor

Full Stack Web Development – AI - Based Note Taking Web App [Github Link](#)

- Developed and **Deployed** a note-taking web application using **React, Python** and **MongoDB** to **vercel**
- Used **API** calls to implement AI features such as **text-to-speech, summarizer, quiz generator** and **note enhancer**
- **Designed** and implemented a modern, simple and user friendly **user interface**

LEADERSHIP AND EXTRACURRICULARS

Director Security and Accommodations LUMS Music Society

June 2024 – June 2025

- Led a team of **10** individuals to manage on-campus accommodations and host **400+** delegates countrywide for the LUMS Music Festival, enhancing their experience and contributing to the event's success

SKILLS & INTERESTS

- **Skills and Frameworks:** Roboflow, ROS, llama.cpp, Git, Docker, Microsoft SQL Server, MongoDB, Flask, Node, React
- **Programming Languages:** C++, Python, C#, SQL, Unity, Matlab, Java, Kotlin
- **Interests:** Music, MMA